

Adversarial Skill Embeddings applications in humanoid robotics

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Outline

- ① Related Work
- ② Adversarial Skill Embeddings
- ③ Applications of The Framework
- ④ Conclusion

Generative Adversarial Architecture

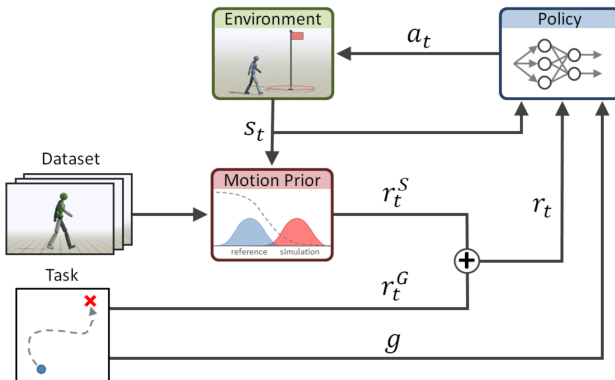


Figure: Adversarial Motion Prior architecture by Peng, Ma, et al. 2021

Training and Rewards

We train simultaneously the discriminator and the policy with:

- **Discriminator:**

$$\min_D -\mathbb{E}_{d\mathcal{M}}[\log D(s_t, s_{t+1})] - \mathbb{E}_{d\pi}[\log(1 - D(s_t, s_{t+1}))]$$

- **Policy:** We define two rewards,

$$r_t^S = -\log(1 - D(s_t, s_{t+1}))$$

$$r_t^G = \text{any goal reward}$$

At the end, the goal is achieved by copying a style dataset.

New task $\Rightarrow$ New training

Mixture of Residual Experts

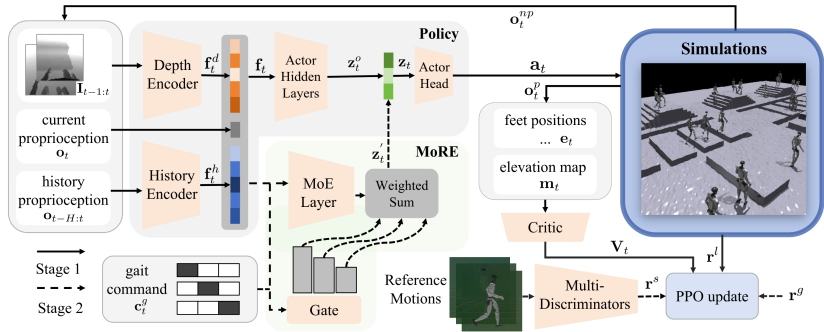


Figure: Mixture of Residual Experts (MoRE) Wang et al. 2025, the gait command chooses an expert policy to use given the states and the camera. The rewards needs to be specified for each gait: $R_t = r_l + \mathbf{1}_{[c_t^g=g]} \left(r_s^{(g)} + r_g^{(g)} \right)$

The Framework

Adversarial Skill Embeddings (ASE) Peng, Guo, et al. 2022 consists of two main components:

- **Low-level policy:** Takes as input the state observations and a latent skill vector $\mathbf{z}$, and produces motions conditioned on $\mathbf{z}$. The reference motion clips are automatically organized in a latent space¹.
- **High-level policy:** Chooses latent skills $\mathbf{z}$ to pass to the low-level policy, learning to sequence and combine them in order to achieve task objectives.

⇒ Finds general transitions between motions

¹A latent space is an embedding of a set of items where similar items are positioned closer together.

Latent space

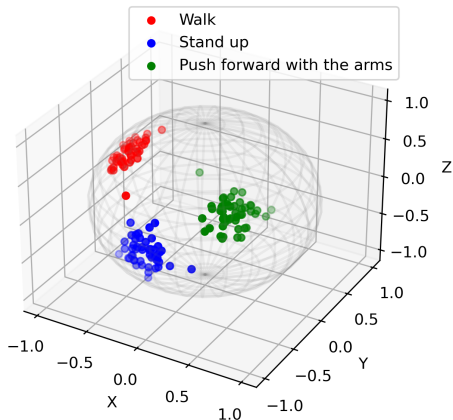


Figure: Illustration of a latent space on a sphere.

Architecture

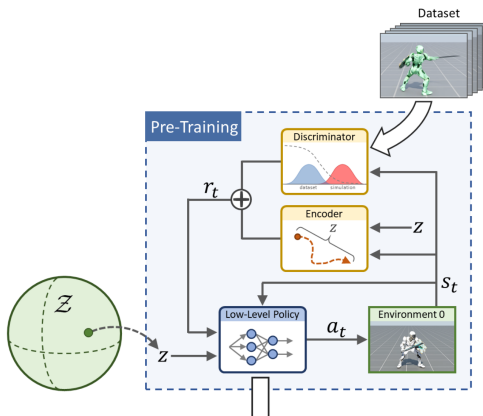


Figure: Architecture of the low-level policy.

Training objective

$$\max_{\pi} \left[\mathbb{E}_{p(\mathbf{Z})} \mathbb{E}_{p(\tau|\pi, \mathbf{Z})} \left[\sum_{t=0}^{T-1} \gamma^t (r_t^S + \beta r_t^E) \right] - w_{\text{div}}, \mathbb{E}_{d^{\pi}(\mathbf{s})}, \mathbb{E}_{\mathbf{z}_1, \mathbf{z}_2 \sim p(\mathbf{z})} \left[\left(\frac{D_{KL}(\pi(\cdot|\mathbf{s}, \mathbf{z}_1), \pi(\cdot|\mathbf{s}, \mathbf{z}_2))}{D_{\mathbf{z}}(\mathbf{z}_1, \mathbf{z}_2)} - 1 \right)^2 \right] \right]$$

The first term is easily recognizable as the cumulative reward and the second term is called the diversity term.

$$r_t^S = -\log(1 - D(s_t, s_{t+1}))$$

$$r_t^E = \kappa \mu_q(s_t, s_{t+1})^\top \mathbf{z}$$

Diversity Loss

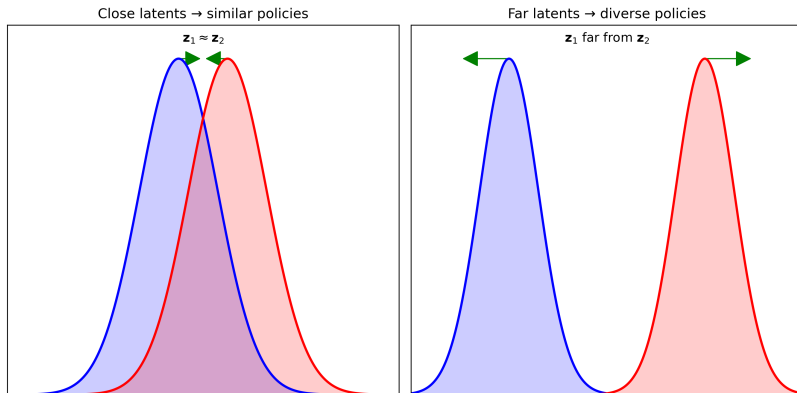


Figure: Illustration of the diversity loss purpose.

High-level Policy

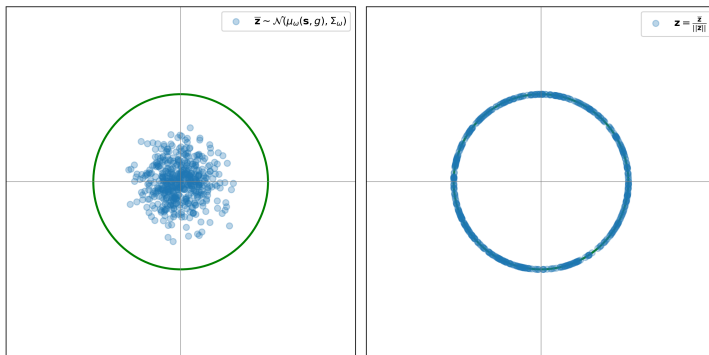


Figure: The high-level policy parameterizes the mean of a Gaussian distribution from which latent variables are sampled. By shifting this mean toward a region of the latent space, the policy effectively selects and activates a specific skill.

High-level Policy

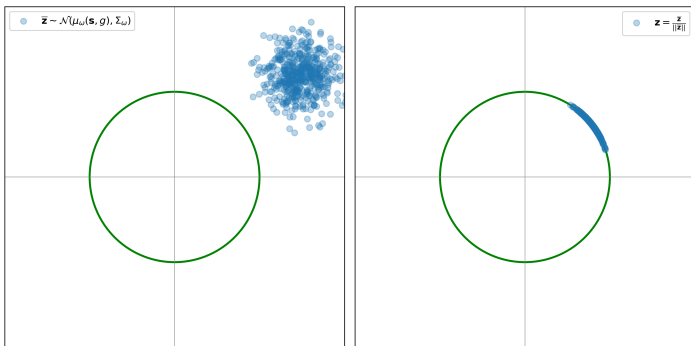


Figure: The high-level policy parameterizes the mean of a Gaussian distribution from which latent variables are sampled. By shifting this mean toward a region of the latent space, the policy effectively selects and activates a specific skill.

Retargeting

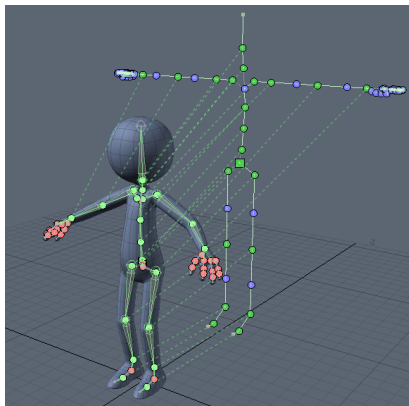


Figure: Retargeting illustration

Designing a Retargeting Pipeline

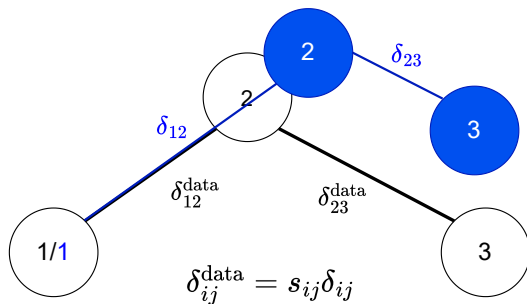


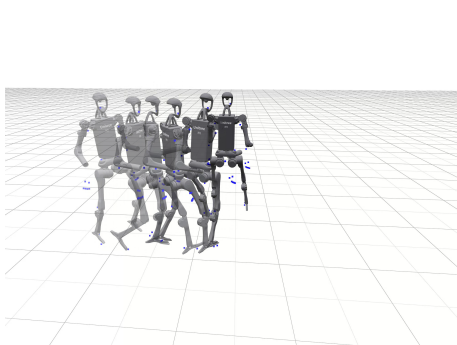
Figure: Alignment between the motion capture skeleton (black) and the robot model (blue). Due to structural differences, an offset exists between corresponding nodes. To achieve optimal retargeting, scaling factors must be computed for each pair of joints to minimize these discrepancies.

Table of Residuals

Residual Name	Formula
Global Alignment	$\sum_i w_i - w_i^{\text{data}}$
Local Alignment Position	$\sum_{(i,j)} \left(\delta_{ij}^{\text{data}} - s_{ij} \delta_{ij} \right)$
Local Alignment Angle	$\sum_{(i,j)} 1 - \cos(\theta_{ij})$
Joint Limit	$\sum_i \max(0, c_i - c_{i,\max}) + \max(0, c_{i,\min} - c_i)$
Rest	$\sum_i c_i - c_{i,\text{rest}}$
Smoothness	$\sum_i c_{i,t} - c_{i,t-1}$, where $c_{i,t}$ and $c_{i,t-1}$ are the i -th joint angle at time t and $t - 1$

Table: Residual terms and their mathematical formulations.

Simple Pipeline and Good Results



Back to the Diversity Term

$$\begin{aligned} \max_{\pi} \mathbb{E}_{p(\mathbf{Z})} \mathbb{E}_{p(\tau|\pi, \mathbf{Z})} \left[\sum_{t=0}^{T-1} \gamma^t (r_t^S + \beta r_t^E) \right] \\ - w_{\text{div}}, \mathbb{E}_{d^{\pi}(\mathbf{s})}, \mathbb{E}_{\mathbf{z}_1, \mathbf{z}_2 \sim p(\mathbf{z})} \left[\left(\frac{D_{KL}(\pi(\cdot|\mathbf{s}, \mathbf{z}_1), \pi(\cdot|\mathbf{s}, \mathbf{z}_2))}{D_{\mathbf{z}}(\mathbf{z}_1, \mathbf{z}_2)} - 1 \right)^2 \right] \end{aligned}$$

To accelerate training, we have to reduce the number of latent dimensions.
When we do so, the diversity term explodes during training $\Rightarrow$ remove it !

Why does the Diversity Term explode?

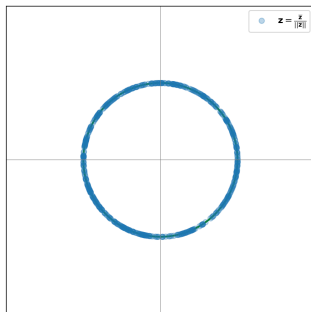


Figure: During the training, latent variables are sampled uniformly on the latent space. The probability of getting two close latent variables decreases as the number of latent dimensions increases. $\mathbb{E}[\langle \mathbf{z}_1, \mathbf{z}_2 \rangle^2] = \frac{1}{d}$.

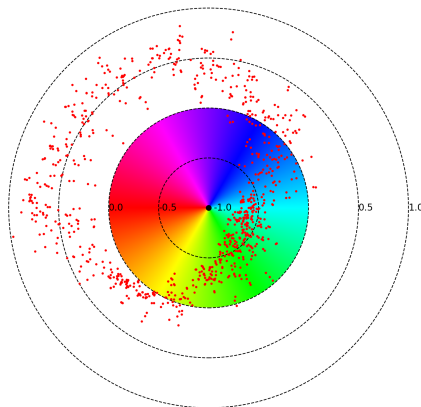
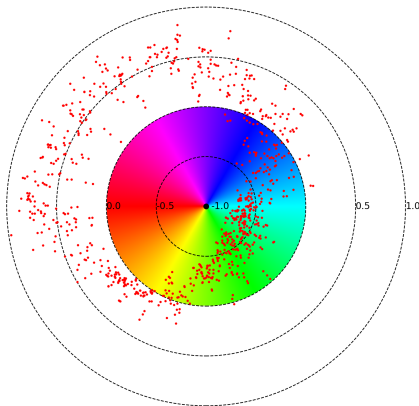
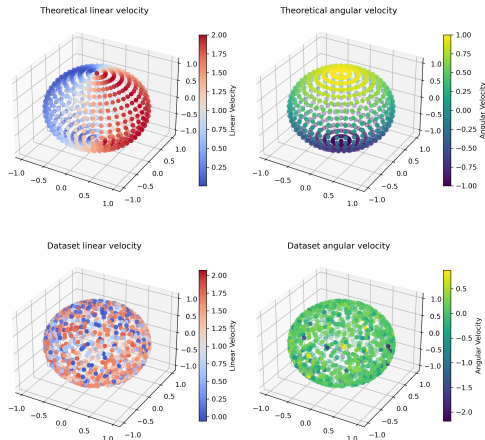


Figure: Color wheel representation of the latent space. The red points indicate measured velocities for sampled latent vectors.

A Simple Experiment: Forward and Backward

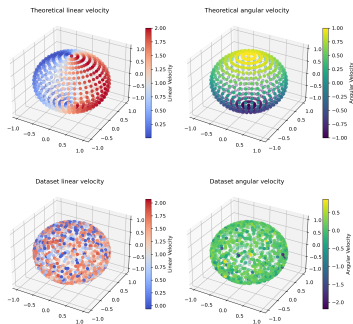


Walking and Turning Policy



Scaling ASE to Complex Robotic Tasks

Walking and Turning Policy



Influence of the Latent Dimension

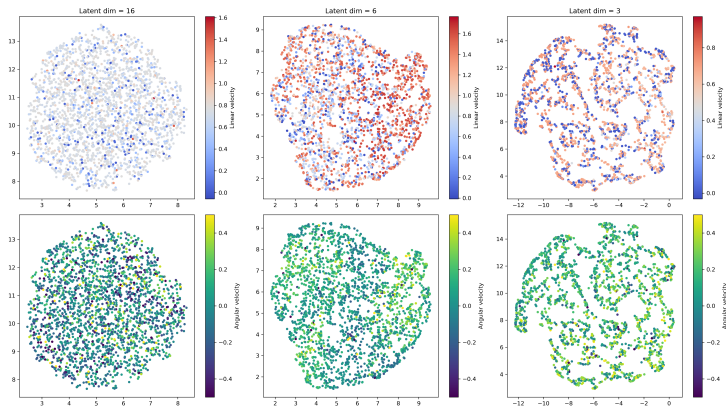
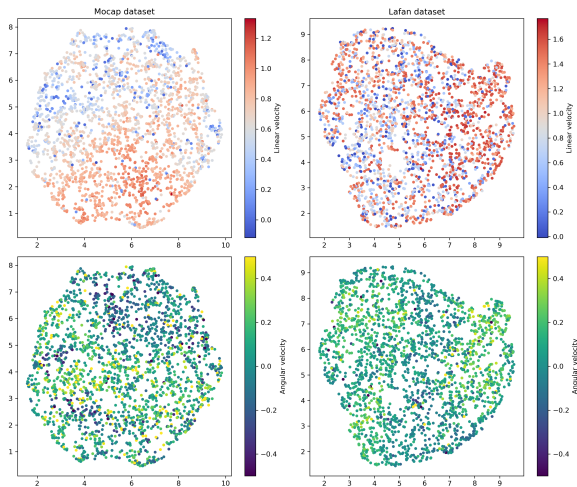


Figure: UMAP projections of latent spaces extracted from three different policies trained on the LAFAN1 dataset.

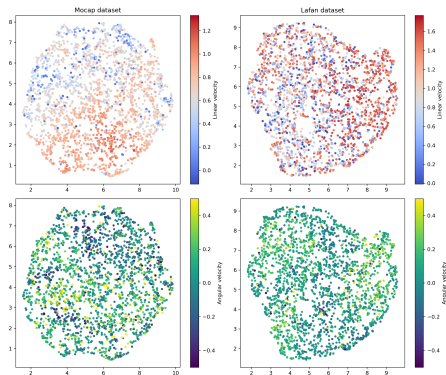
Scaling ASE to Complex Robotic Tasks

Influence of the Dataset



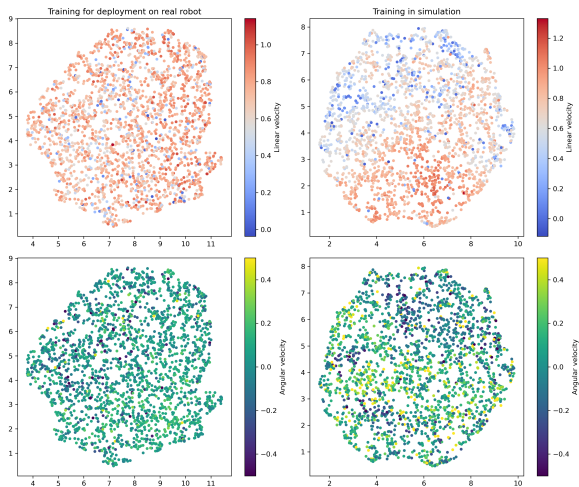
Scaling ASE to Complex Robotic Tasks

Influence of the Dataset

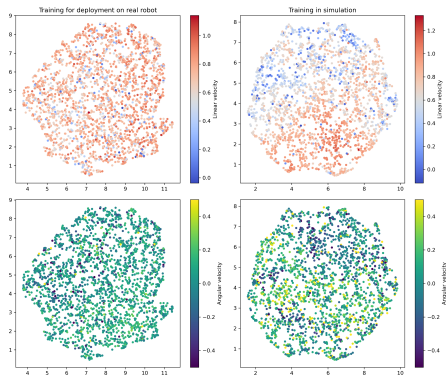


ASE on Real H1

Influence of the Sim-to-Real Gap



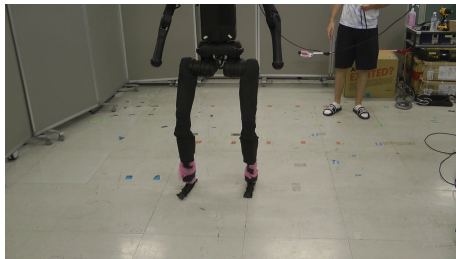
Influence of the Sim-to-Real Gap



Human-like Walking on the Real H1 Robot



Human-like Walking on the Real H1 Robot



Applying ASE to a Door Opening Task

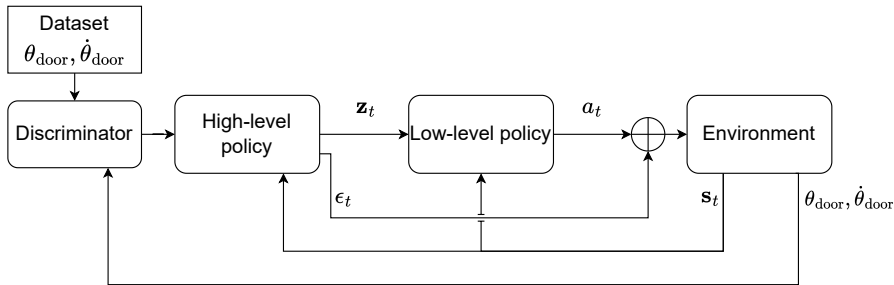
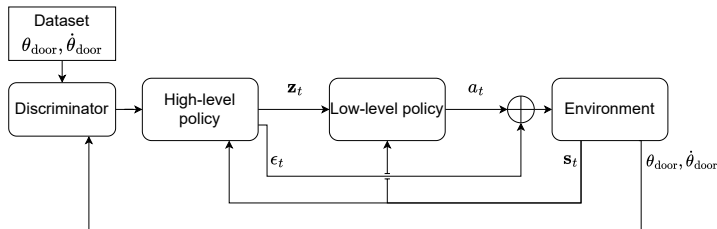


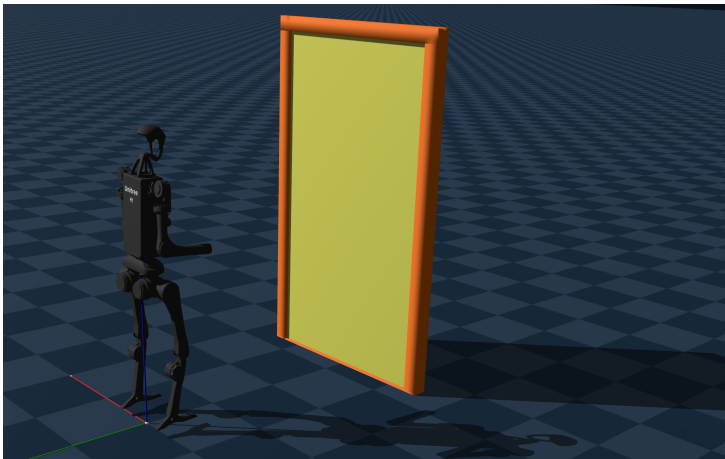
Figure: Modified ASE pipeline for the door-opening task. The high-level policy generates latent variables $\mathbf{z}_t$ for the low-level controller, which outputs actions a_t along with a residual correction ϵ_t . The final executed action is $a_t + \epsilon_t$. A door-specific discriminator provides imitation rewards by comparing the simulated motion with Mocap demonstrations of door opening.



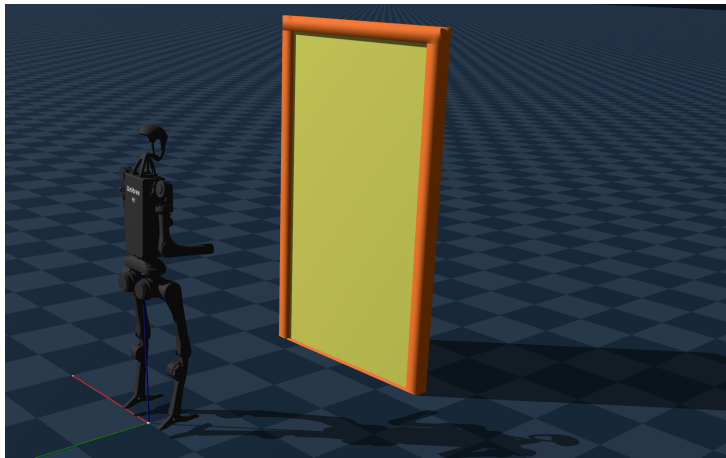
Applying ASE to a Door Opening Task



Classical Imitation Pipeline



Classical Imitation Pipeline



Conclusion and Perspectives

- Applied **Adversarial Skill Embeddings** to humanoid locomotion with the Unitree H1 robot.
- Investigated:
 - Latent dimensionality and skill representation
 - Role of diversity loss
 - Retargeting pipeline for motion capture datasets
- ASE produces smooth, human-like walking transitions, but is sensitive to noise and lacks scalability.
- **Diffusion-based policies** Liao et al. 2025 surpass ASE in robustness and multi-skill generalization.
- Promising directions: **Forward-Backward Representations** Touati and Ollivier 2021 for generalizable multi-skill locomotion policies.

References I



Liao, Qiayuan et al. (2025). *BeyondMimic: From Motion Tracking to Versatile Humanoid Control via Guided Diffusion*. arXiv: 2508.08241 [cs.R0]. URL: <https://arxiv.org/abs/2508.08241>.



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References II



Touati, Ahmed and Yann Ollivier (2021). *Learning One Representation to Optimize All Rewards*. arXiv: 2103.07945 [cs.LG]. URL: <https://arxiv.org/abs/2103.07945>.



Wang, Dewei et al. (2025). *MoRE: Mixture of Residual Experts for Humanoid Lifelike Gaits Learning on Complex Terrains*. arXiv: 2506.08840 [cs.RO]. URL: <https://arxiv.org/abs/2506.08840>.